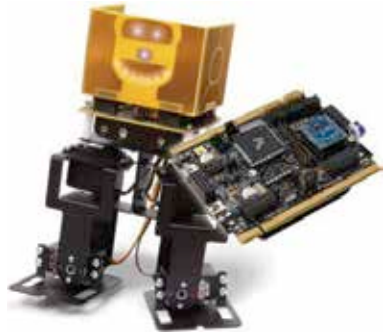


with a Basic interpreter which can run the Basic code. The programming is very easy too, consider the following code used to read the accelerometer values from an MMA7455 accelerometer on a Freescale Bipedal Mechatronics Robot based on the 32-bit ColdFire MCF52259. This runs a version of StickOS which comes with a Basic interpreter:



Freescale Mechatronics Robo running StickOS

```

5 rem tilt_demo
10 dim x, y, z
20 gosub mma7455_init 0x1d // library function to initialise
// MMA7455
30 while 1 do // infinite loop
40 gosub mma7455_poll 0x1d, x, y, z // library function to read
// MMA7455
50 print hex "x = ", x, "y = ", y, "z = ", z
60 sleep 100 ms
70 endwhile
  
```

## Building Software

This is the most crucial section of this chapter. In this section we'll discuss how to run the software that you've written on the target system. This part applies to the programs written in C and Ada. The other programming languages that we mentioned are interpreted except Assembly which is directly converted to machine code. The usual workflow when developing for an embedded system is:

- Create a C file(s) on the host which contain all your functions.
- Compile/Assemble the machine code on the host
- Link all the object files and libraries and resolve all symbols on host
- Assign memory addresses to the code and data on host
- Download the executable image onto the target processor memory
- Reset the target processor
- Voila! Your program is working.